DEPARTMENT OF METEOROLOGY

Lab Assignment 02

COURSE TITLE

Web GIS Development

SUBMITTED BY

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SUBMITTED TO

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ENVIRONMENTAL MONITORING AND MANAGEMENT WEB GIS

Environmental Monitoring Pakistan

- **Project Information:**

- **Project Name: Environmental Monitoring Pakistan**
- **Topic: Environmental Monitoring and Station Management System**
- **Study Area: Pakistan (Nationwide)**

1. Introduction

Environmental Monitoring Pakistan is a database-driven Web GIS application developed to display and manage environmental monitoring stations across Pakistan.

The website allows users to view station locations, search and filter records, and add or update station information. All station data is stored in a PostgreSQL database with the PostGIS extension.

The system connects PostgreSQL/PostGIS, Node.js, Express.js, and Leaflet to provide an interactive mapping application.

2. Objectives

The main objectives of this project are:

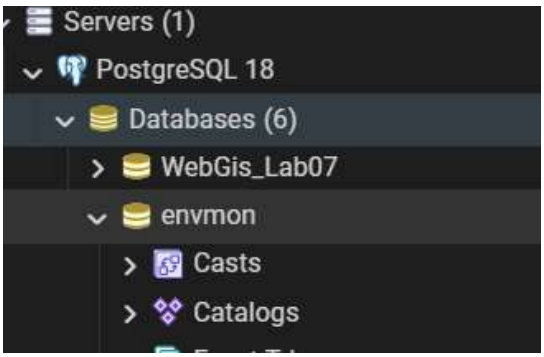
- Create a spatial database using PostgreSQL and PostGIS.
- Store at least 20 environmental monitoring stations.
- Perform SQL operations on database records.
- Connect the database with a Node.js and Express backend.
- Display station locations on a Leaflet map.
- Implement searching, filtering, adding, and updating records.
- Verify data persistence and input validation.

3. Tools and Technologies

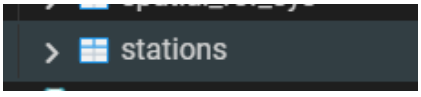
Technology	Purpose
PostgreSQL	Store station and observation records
PostGIS	Store and process spatial geometry
pgAdmin 4	Manage database and execute SQL
Node.js	Run the backend server
Express.js	Create REST API endpoints
HTML, CSS, JavaScript	Develop the website interface
Leaflet.js	Display interactive maps
OpenStreetMap and Esri	Provide basemap layers
Visual Studio Code	Write and manage project files

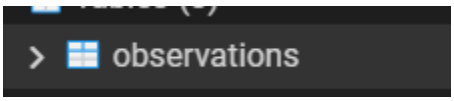
4. Database Creation

A PostgreSQL database was created and the PostGIS extension was enabled to support spatial data.



The database contains two tables:

1. Stations: Stores station names, types, provinces, cities, status, data sources, elevations, and point geometry. 

2. Observations: Stores rainfall, temperature, PM2.5 measurements, and observation dates. Each observation is linked to a station through station_id. 

The stations table uses geometry(Point, 4326) to store geographic coordinates in WGS 84.

5. Spatial Data Population

The database was initially populated with 25 sample environmental monitoring stations across Pakistan.

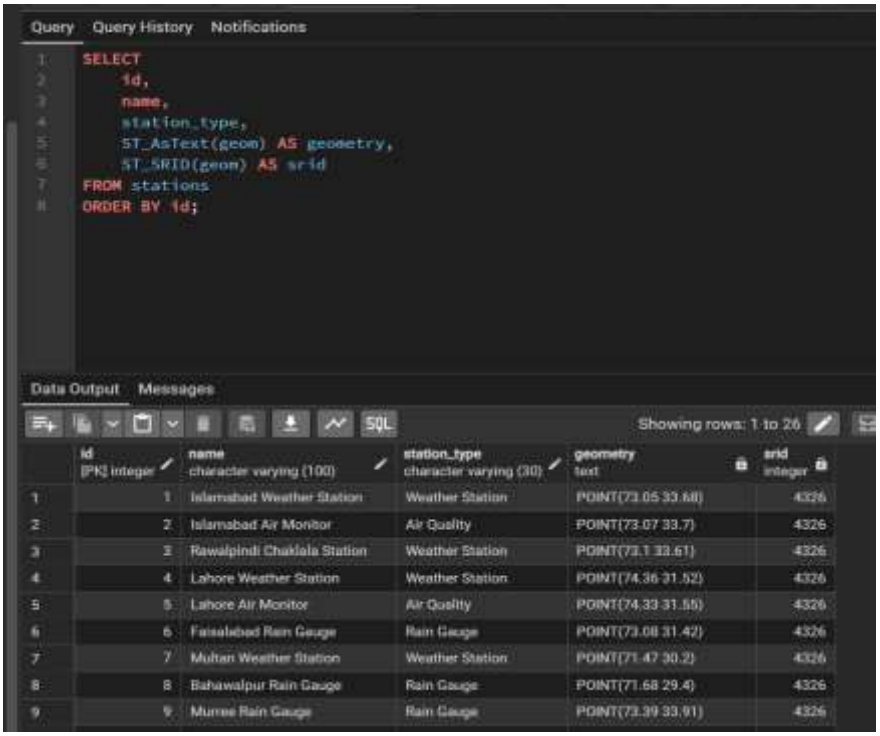
The station types include:

- Weather Stations
- Rain Gauges
- Air Quality Monitors
- River Gauges

Each station contains a point location with latitude and longitude. The application also stores environmental observations in a related table.

The initial station locations are approximate, and the generated observation values are sample data for demonstration.

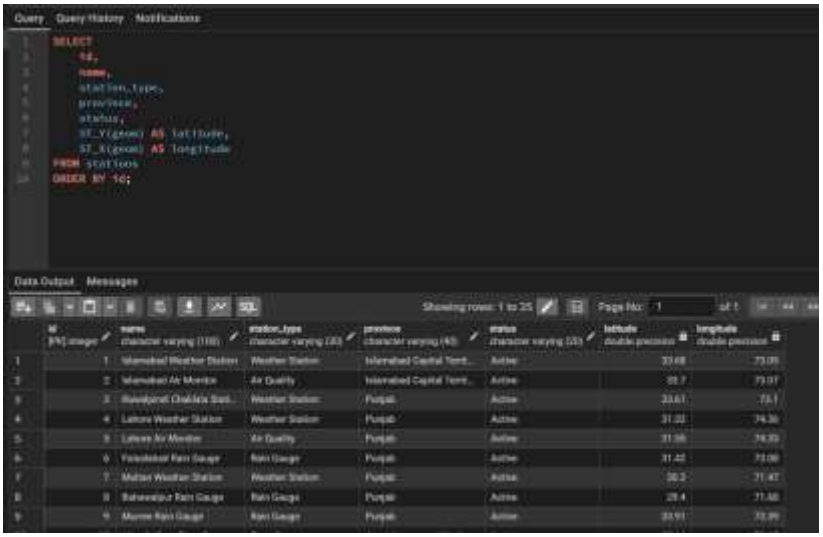
SQL query:



6. SQL Operations

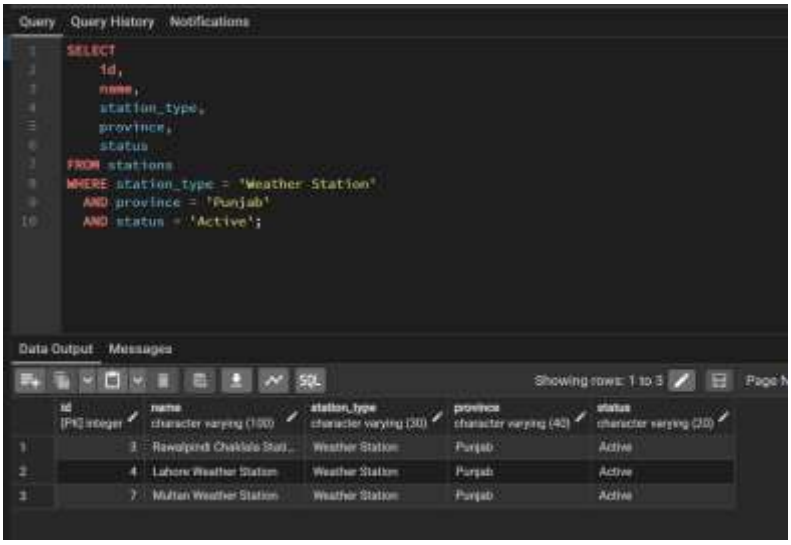
Different SQL queries were performed to retrieve, filter, count, and update database records.

6.1 Retrieve Record:



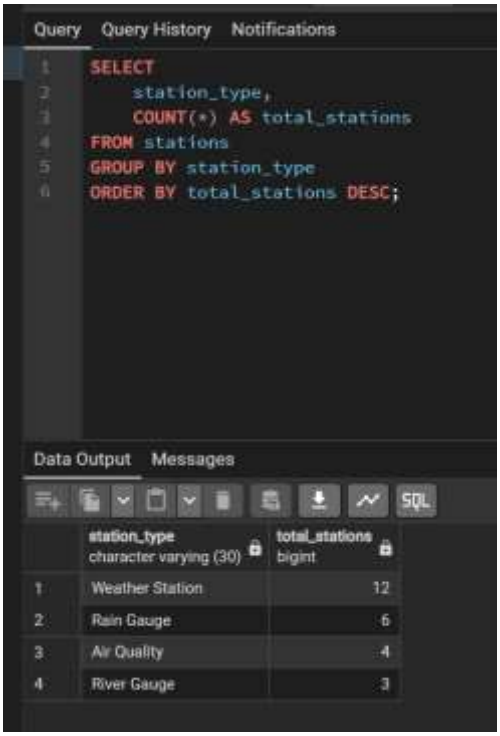
This query retrieves monitoring stations from the database.

6.2 Filter Records



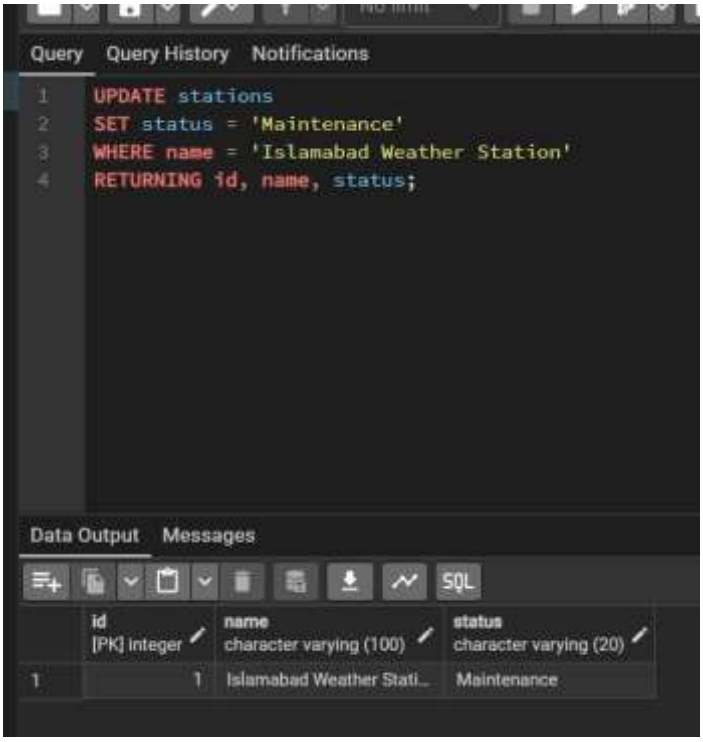
This query filters active weather stations located in Punjab.

6.3 Count Records



This query counts the stations belonging to each monitoring category.

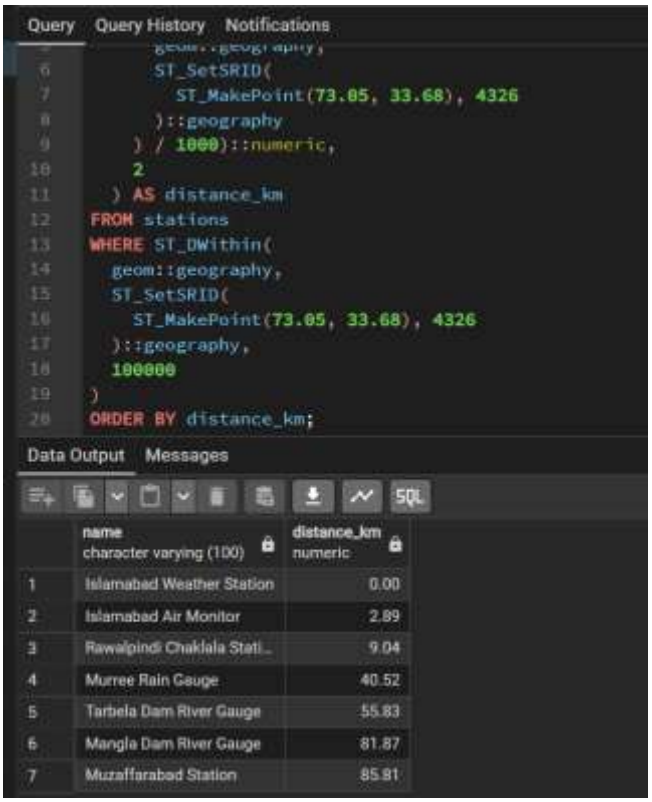
6.4 Update Records



This query updates the station status in PostgreSQL.

6.5 Spatial Query

The PostGIS function ST_DWithin() was used to identify stations within 100 km of Islamabad.



This demonstrates spatial distance analysis using PostGIS.

7. Backend Development and API

The backend was developed using Node.js and Express.js.

The Express server connects to PostgreSQL using the pg library and retrieves records through SQL queries.

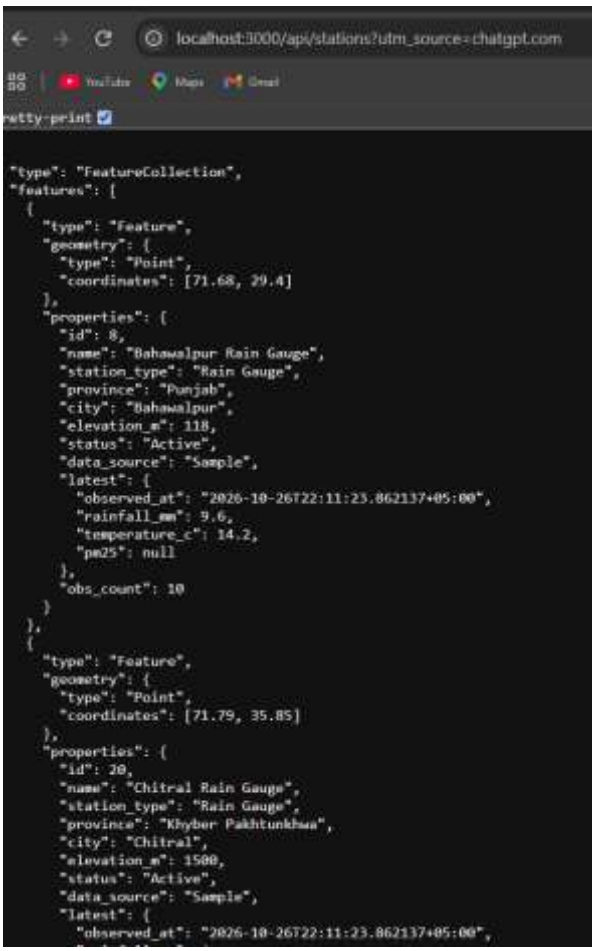
The API returns station locations and attributes in GeoJSON format, which can be displayed directly using Leaflet.

Main API endpoints:

API endpoint	Function
GET /api/health	Check database connection

GET /api/stations	Retrieve station records
GET /api/provinces	Retrieve provinces
GET /api/stats	Retrieve station statistics
POST /api/stations	Add a new station
PUT /api/stations/:id	Update an existing station
POST /api/stations/:id/observations	Add an observation

The backend also uses parameterized SQL queries and authentication to protect database modifications.



8. Web GIS Development

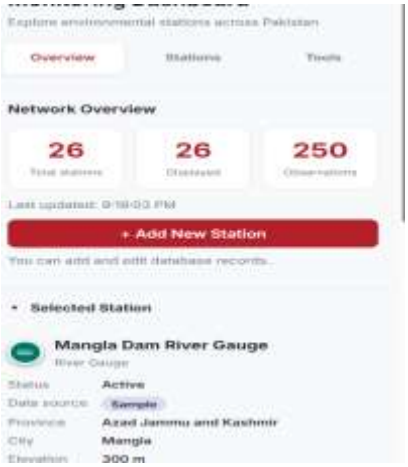
The frontend was developed using HTML, CSS, JavaScript, and Leaflet.js.

Leaflet retrieves station data from the backend API and displays monitoring stations as markers on an interactive map.

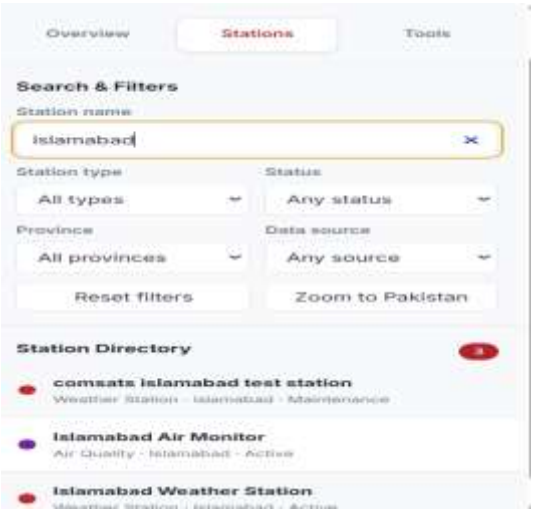


The website includes three sidebar sections:

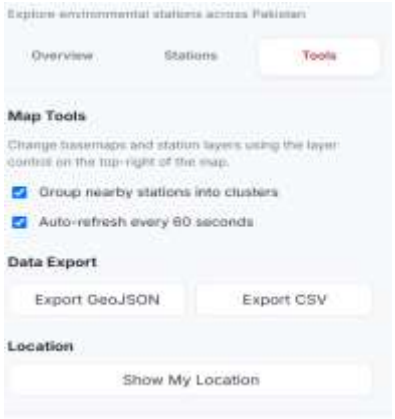
- **Overview**



- **Stations**



- **Tools**.



Users can select a marker to view station details, observations, coordinates, and status.

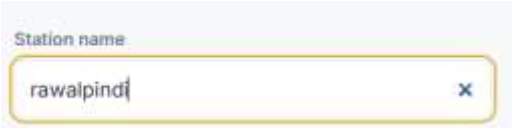


9. Search and Filtering

The website provides searching and filtering using multiple station attributes.

Users can filter records by:

- **Station name**



- **Station type**



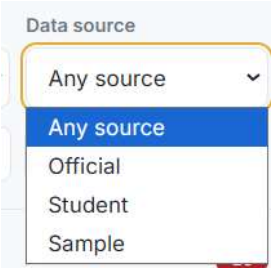
- **Province**



- **Station status**



- **Data source**



The filtered results are retrieved through the backend API and displayed on the map and station directory.

10. Record Creation and Updating

The website includes forms for adding new stations and editing existing stations.

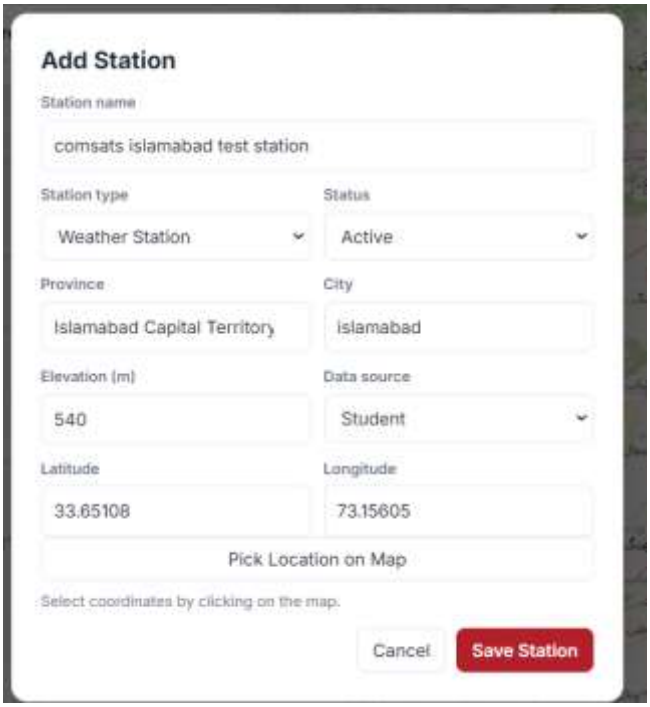
To test this functionality, a new station named **COMSATS Islamabad Test Station** was created through the website.

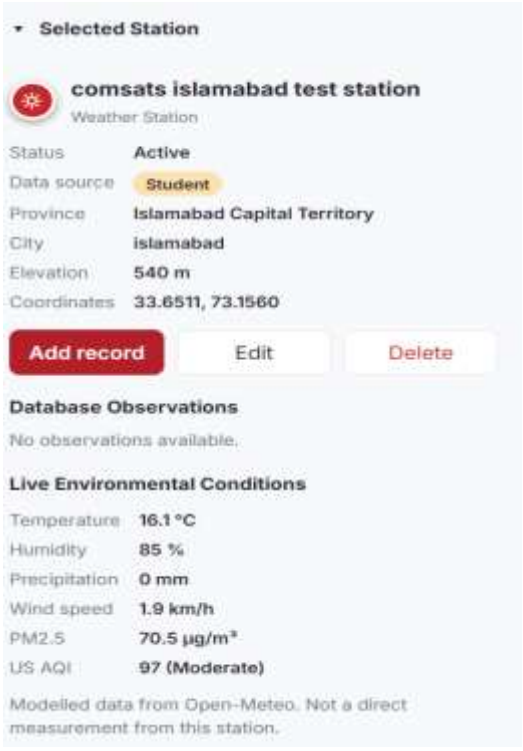
The form included station type, name, province, city, status, elevation, and geographic coordinates.

The record was successfully saved and verified in pgAdmin.

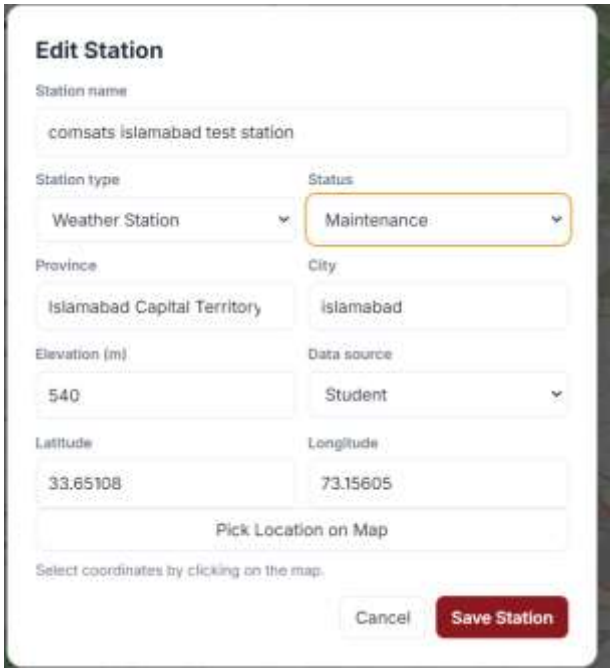
The station status was then changed from Active to Maintenance using the Edit Station form.

Add Station:





Edit Station:



11. Data Persistence

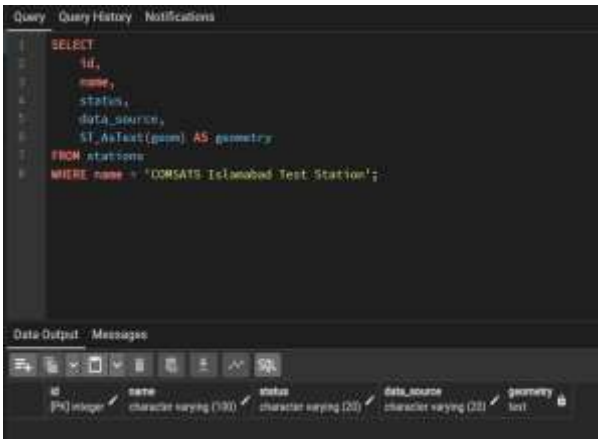
Data persistence was tested by updating the test station through the website.

After saving the changes, the updated record was checked in PostgreSQL using pgAdmin.

The website was refreshed, and the updated station information remained available.

This confirmed that changes were stored in PostgreSQL rather than only in the browser.

SQL verification:





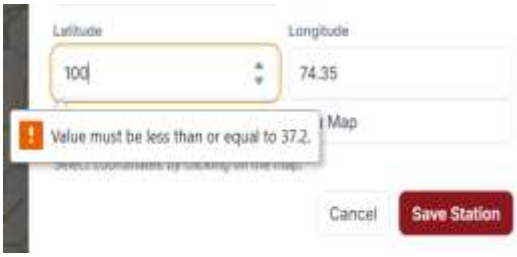
12. Input Validation and Error Handling

Input validation was implemented in both the website forms and backend.

For example, latitude values must be between 23.5 and 37.2, while longitude values must be between 60.8 and 77.9.

Required fields cannot be left empty, and invalid measurements are rejected.

API errors and database connection issues are also handled by the application.

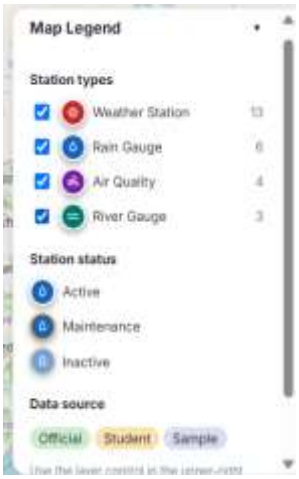


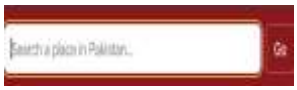
13. Additional Features

The application includes several features beyond the basic assignment requirements:

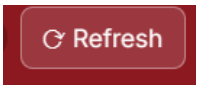
- Two basemaps: OpenStreetMap and Esri Satellite. 

- Layer controls to show or hide station categories. 

- Collapsible map legend. 

- Search for places across Pakistan. 



- Automatic data refreshing. 

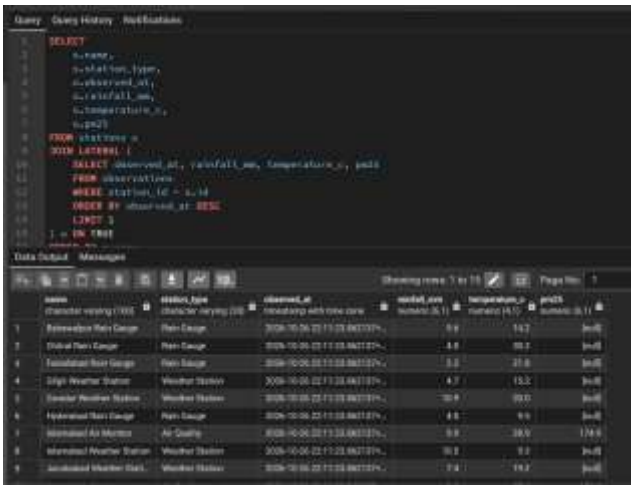
- Current-location feature. 

- Export station records as CSV and GeoJSON. 

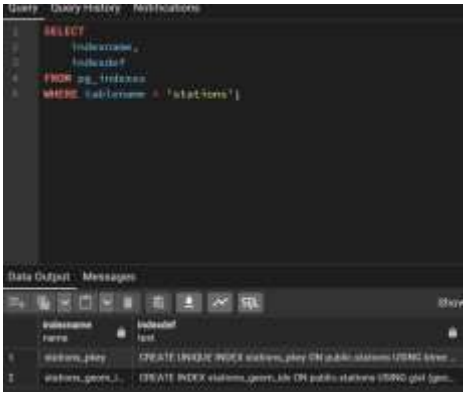
- Live modelled weather and air-quality information from Open-Meteo 

- Login authentication for database modifications. 

- Related observations table.



- PostGIS spatial distance queries and GiST spatial indexing.



- Responsive interface with light and dark themes.





14. Conclusion

The Environmental Monitoring Pakistan Web GIS application was successfully developed using PostgreSQL/PostGIS, Node.js, Express.js, and Leaflet.

The system stores spatial station records, displays them on an interactive map, and supports searching, filtering, adding, and updating data.

Database persistence and input validation were also tested successfully.

The project demonstrates how spatial databases and web technologies can be integrated to develop a dynamic Web GIS application.

GitHub Repository: https://github.com/HafsaAmjad29/Environmental_Monitoring_Pakistan

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